# CFD Analysis of Automobile Radiator

Focus: Copper Material Case with Low-Velocity Operation

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## 1 Introduction

The radiator in an automobile plays a key role in controlling engine temperature by transferring heat from the coolant to the surrounding air. When the system fails to remove heat effectively, the engine can overheat, leading to reduced performance and possible damage. To understand how efficiently a radiator performs this heat exchange, numerical simulations using Computational Fluid Dynamics (CFD) provide a detailed and cost-effective approach.

This project focuses on analyzing the heat transfer and flow behavior inside an automobile radiator using **ANSYS Fluent**. Instead of performing a full experimental study, a simplified radiator geometry was modeled, meshed, and tested virtually. The simulation helps visualize temperature gradients, pressure variations, and surface heat flux, giving insights into how design parameters affect cooling performance.

## 2 Objectives

The main goals of this study are:

- To create a 3D model of an automobile radiator core with realistic geometry.
- To apply suitable mesh refinement and boundary conditions for reliable CFD results.
- To simulate transient fluid flow and heat transfer using the $k$–$\epsilon$ turbulence model.
- To analyze pressure drop, temperature distribution, and wall heat flux.
- To evaluate overall cooling performance and identify potential design improvements.

## 3 Methodology

The overall workflow followed the standard ANSYS process: ANSYS Workbench $\rightarrow$ DesignModeler (Geometry) $\rightarrow$ Mechanical model (Meshing) $\rightarrow$ Fluent (Setup, Solution, and Results).

### 3.1 Geometry

The radiator was modeled as a rectangular box representing the core with 16 evenly spaced copper tubes. The dimensions were:

- Radiator box: $250\,\mathrm{mm} \times 290\,\mathrm{mm} \times 44\,\mathrm{mm}$
- Tube radius: $6\,\mathrm{mm}$
- Number of tubes: 16

Copper was selected for both the tubes and fins because of its high thermal conductivity and common industrial use.

### 3.2 Meshing

A fine mesh was generated using ANSYS Meshing, with a minimum element size of $5.e - 0.003\,\mathrm{m}$. The region near the tube walls was refined to capture boundary layer effects more accurately. Mesh quality checks confirmed that skewness and orthogonal quality were within acceptable limits.

### 3.3 Solver Setup

The simulation was performed in **ANSYS Fluent**. The setup included:

- Fluid: Water (liquid)
- Inlet velocity: $0.5\,\mathrm{m/s}$
- Inlet temperature: $353\,\mathrm{K}$
- Material: Copper (tubes and fins)
- Fin heat transfer coefficient: $120\,\mathrm{W/m^2K}$
- Turbulence model: $k$–$\epsilon$ with enhanced wall treatment
- Gravity: $-9.81\,\mathrm{m/s^2}$ in Y-direction
- Simulation type: Transient (100 time steps)

At the inlet, a velocity boundary condition was applied, and the outlet was set to atmospheric pressure. Walls were defined as no-slip boundaries with appropriate heat transfer conditions.

## 4 Post-processing and Figures

The post-processing stage involved visualizing residual plots, temperature contours, and pressure distributions to understand how heat and momentum were transferred through the radiator. The main results are shown below.

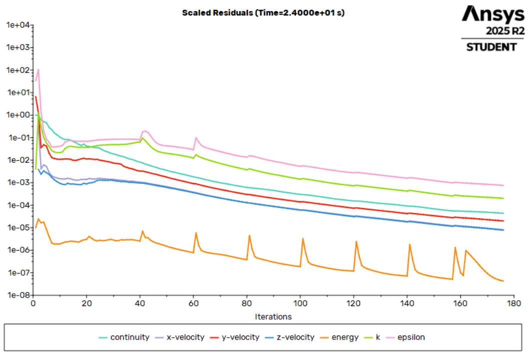


Figure 1: Convergence History of Scaled Residuals in Automobile Radiator Simulation

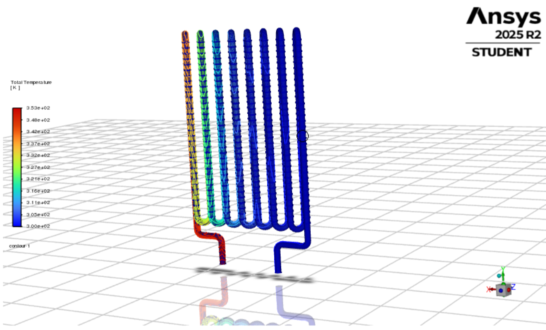


Figure 2: Total Temperature Distribution in Automobile Radiator (Side View)

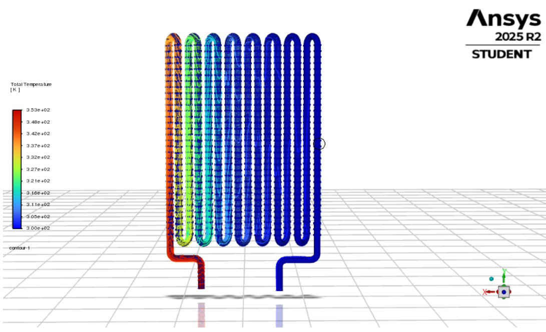


Figure 3: Total Temperature Distribution in Automobile Radiator (Front View)

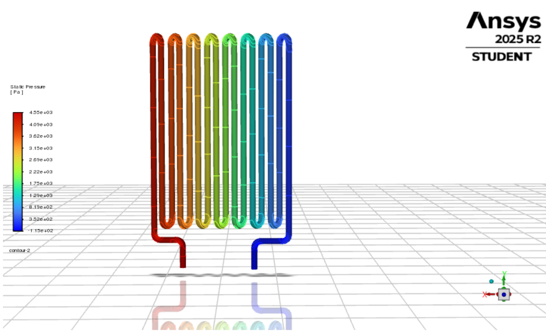


Figure 4: Static Pressure Distribution Along Automobile Radiator Tubes

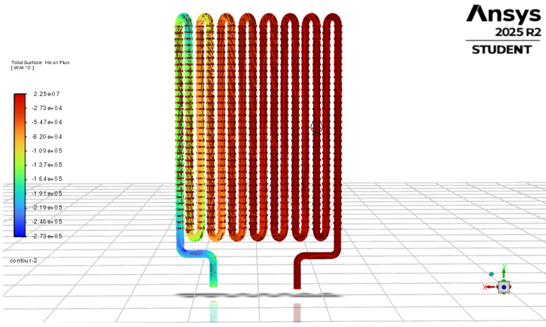


Figure 5: Total Surface Heat Flux on Automobile Radiator Wall

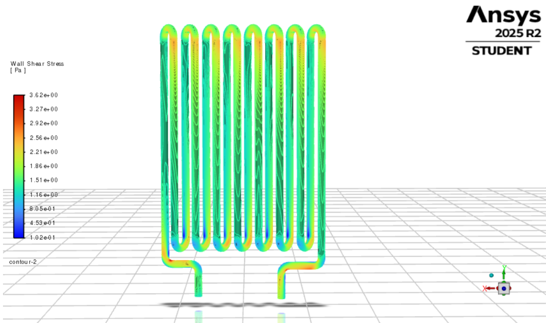


Figure 6: Total Wall Shear Stress Distribution in Automobile Radiator

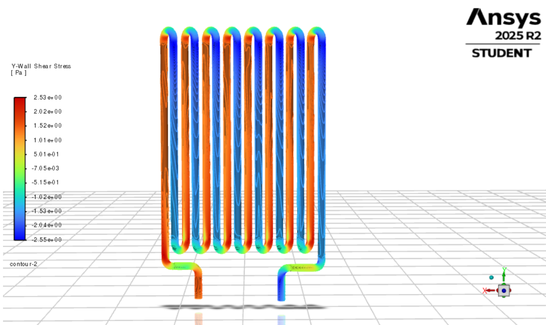


Figure 7: Y-Direction Wall Shear Stress in Automobile Radiator (Secondary Flow Indicator)

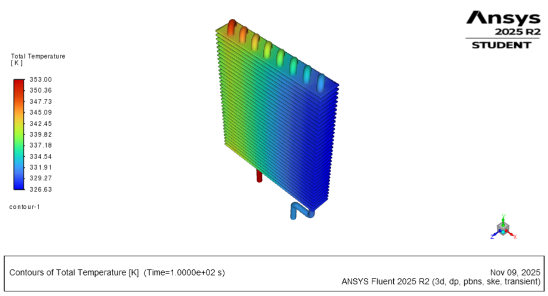


Figure 8: Total Temperature Contours in Automobile Radiator at $t = 100\,s$ (Isometric View)

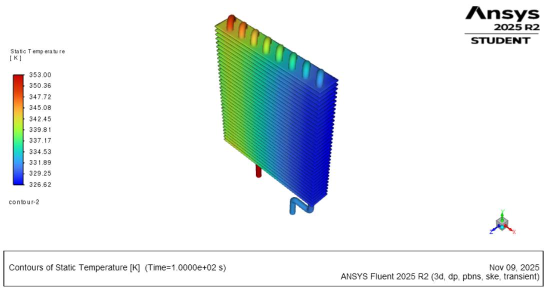


Figure 9: Static Temperature Contours in Automobile Radiator at $t = 100\,s$ (Isometric View)

## 5 Results and Discussion

The temperature contours clearly showed a gradual decrease in coolant temperature as it flowed through the tubes. The cooling was most effective near the fin regions, where the heat transfer coefficient played a major role in extracting heat.

A noticeable pressure drop occurred between the inlet and outlet, which is expected due to viscous effects and flow resistance within narrow tubes. The $k$–$\epsilon$ turbulence model with enhanced wall treatment gave stable results and captured near-wall effects reasonably well for this setup.

The wall shear stress distribution indicated regions of secondary flow and localized turbulence that slightly enhanced mixing. These small-scale flow structures are beneficial for improving local heat transfer, although they may also contribute to pressure loss.

Overall, the radiator performed as expected under the chosen conditions , effectively reducing coolant temperature while maintaining stable flow characteristics. The residuals converged smoothly, suggesting that the numerical setup was consistent.

## 6 Conclusion

This project successfully modeled and simulated an automobile radiator using ANSYS Fluent. The results showed effective cooling of the fluid, with clear temperature gradients and pressure drops across the radiator. The study confirmed that CFD can provide valuable insight into how geometric design and boundary conditions influence performance.

While the model was simplified, it still captured the essential heat transfer mechanisms. With more detailed geometry and air-side coupling, the predictions could be even closer to experimental data.

## 7 References

1. A. S. Patil et al., "Analysis of Automobile Radiator using Computational Fluid Dynamics," *ResearchGate*, 2020.

2. ANSYS Fluent Theory Guide, Release 2023 R1.

# Collaboration and Workflow

All tasks were carried out collaboratively by the team members.

Table 1: Task Distribution and Outputs

| Step | Task | Person | Output |
|---|---|---|---|
| 1 | Geometry Creation | Prakrut | Cleaned 3D radiator model |
| 2 | Meshing & Boundary Setup | Anagha | Quality mesh + named selections |
| 3 | Simulation & Post-processing | Aarav | Fluent results + report |

**Resources and Repositories:**

- **GitHub Repository:** github.com/your-repo-link
- **Google Drive Folder:** drive.google.com/your-drive-link